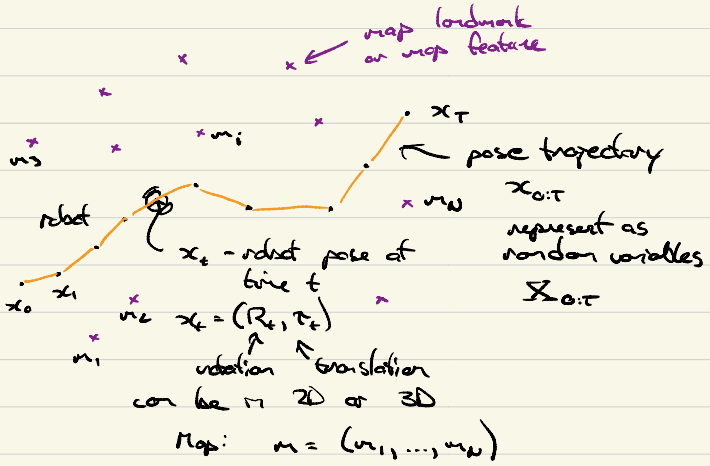
Lecture 14

- SLAM, Factor Graphs
- Rigid Body Poses
- Example Factors

Recall SLAM:



Map: $m = (m_1, \ldots, m_N)$

can model as random variables

$M = (M_1, \ldots, M_N)$

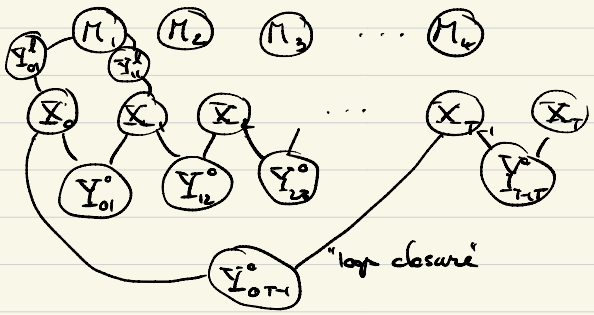
Typically we estimate $(\mathbf{X}_{0:T}, M)$ with a MAP estimate:

$$(x_{0:T}^{MAP}, m^{MAP}) = \underset{x_{0:T}, m}{\operatorname{argmax}} \sum_{t=1}^{T} \left(\sum_{i \in L_t} (\text{landmark factor}) + (\text{odometry factor}) \right)$$

L_t: set of landmark visible at pose x_t

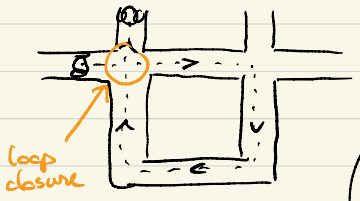
Factor Graphs

First Consider Probabilistic Graphical Model for SLAM

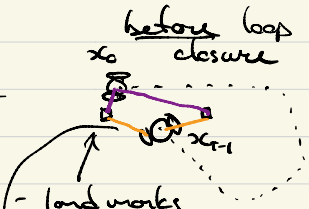


E.g. Importance of loop closure

Ground Truth:

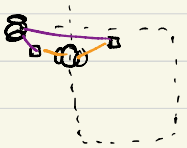


before loop closure



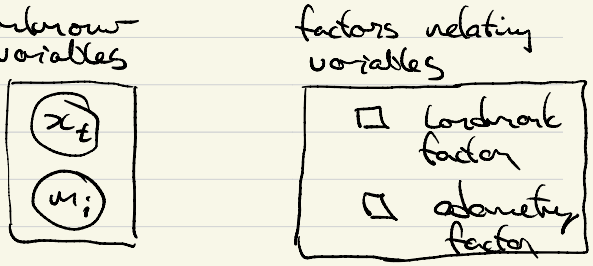
- landmarks stored in memory
- at a later time robot recognize the same landmarks it has seen before
- can back out a relative pose between x_0 and x_{T-1}
- call it an odometry measurement $Y_{0,T-1}^o$

after loop closure



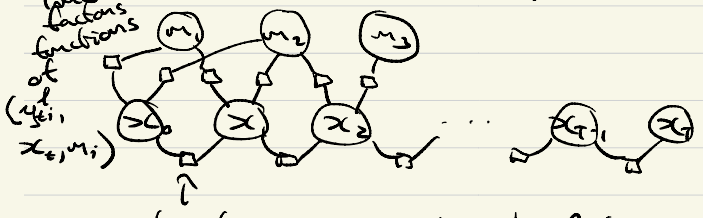
- we get a global correction

Factor Graph Formalism for SLAM



factors are function relationships between variable implied by measurements, prior info, or motion models

SLAM Factor Graph



odometry factors, function of odometry measurements and poses $(y^o_{t-1t}, x_{t-1}, x_t)$

- typically factors come from a measurement model
- E.g. odometry

$$y^o_{t-1t} = g_o(x_{t-1}, x_t) + v^o_{t-1t}$$

where $v^o_{t-1t} \overset{\text{noise}}{\sim} N(0, \Sigma_o)$

so the corresponding factor would be:

high-level template for odometry factor

$$\boxed{\| y^o_{t-1t} - g_o(x_{t-1}, x_t) \|^2_{\Sigma_o^{-1}}}$$

this is $-\log$ of the cond. dist. "$p(y|x)$" we saw last time.

- E.g. Landmark factor

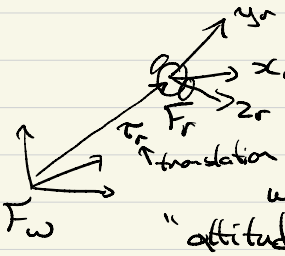
$$y^l_{ti} = g_l(x_t, m_i) + v^l_{ti}$$

where $v^l_{ti} \sim N(0, \Sigma_l)$

so the factor is:

$$\boxed{\| y^l_{ti} - g_l(x_t, m_i) \|^2_{\Sigma_l^{-1}}}$$

Rigid Body Poses



body fix coord. axes expressed in world Frore F_w

we define robots "attitude" or "orientation" as

$$R = \left[\begin{matrix} | & | & | \\ x_r & y_r & z_r \\ | & | & | \end{matrix} \right] \in SO(3)$$

mutually orthogonal unit vectors

recall orthogonal matrices are defined by

all equivalent:
- (i) mutually orthogonal unit vector columns
- (ii) $R^T R = I$, $RR^T = I$
- (iii) $R^T = R^{-1}$ inversion

we say $R \in O(d)$

rotation matrices are "Special" Orthogonal matrices because

$$\det(R) = +1$$

(not -1)

R defines rigid body rotations

Q: why not $\det(R) = -1$? what are these matrices?

A: These are chiral reflections (e.g. right hand $\Leftrightarrow$ left hand)

robot pose $x_t = (R_t, \tau_t)$

τ_t: translation in $\mathbb{R}^d$, $d = 2$ or 3

$(R_t, \tau_t) \in SE(d)$ $(d = 2 \text{ or } 3)$

Special Euclidean

all rigid body transforms

Intuitive Meaning

transforms a vector seeing in the robots frame to the world frame:

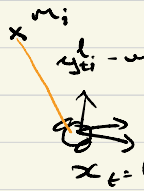
let v_r – vector in robot frame

same vector in world frame

$$\boxed{v_w = R_t v_r + \tau_t}$$

Consider Specific Examples of Landmark and Odometry Factors:

E.g. landmark factors



suppose measurement is relative position of landmark w.r.t x_t

$$m_i = R_t y^l_{ti} + \tau_t + v^l_{ti}$$

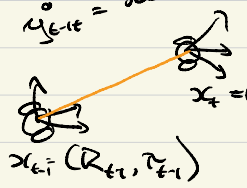
m_i: world frame

$v^l_{ti} \sim N(0, \Sigma_l)$

landmark factor: $\boxed{\| m_i - R_t y^l_{ti} - \tau_t \|^2_{\Sigma_l^{-1}}}$

E.g. Odometry factor

u^o_{t-1t} = relative pose of x_t w.r.t. x_{t-1}



measured quantity

$$u^o_{t-1t} = (\hat{R}_{t-1t}, \hat{\tau}_{t-1t})$$

Let's consider vector v_t in frame x_t

how to get v_{t-1} in frame x_{t-1}?

world frame

$$v_w = R_t v_t + \tau_t$$
$$v_w = R_{t-1} v_{t-1} + \tau_t$$

$$R_t v_t + \tau_t = R_{t-1} v_{t-1} + \tau_{t-1}$$

$$v_{t-1} = R_{t-1}^T (R_t v_t + \tau_t - \tau_{t-1})$$

$$v_{t-1} = \underbrace{(R_{t-1}^T R_t)}_{\text{relative rotation between } x_t, x_{t-1}} v_t + \underbrace{R_{t-1}^T (\tau_t - \tau_{t-1})}_{\text{relative translation of } x_t \text{ w.r.t. } x_{t-1}}$$

we get two quadratic terms:

rotation –

$$\hat{R}_{t-1t} \approx R_{t-1}^T R_t$$

$$\Rightarrow \quad R_{t-1} \hat{R}_{t-1t} \approx R_t$$

$$R_{t-1} \hat{R}_{t-1t} = R_t + v^o_{t-1t}$$

$$v^o_{t-1t} \sim N(0, \Sigma_R)$$

rotation factor:

$$\boxed{\| R_{t-1} \hat{R}_{t-1t} - R_t \|^2_{\Sigma_R^{-1}}}$$

translation –

$$\hat{\tau}_{t-1t} \approx R_{t-1}^T (\tau_t - \tau_{t-1})$$

translation factor:

$$\Rightarrow \boxed{\| R_{t-1} \hat{\tau}_{t-1t} - (\tau_t - \tau_{t-1}) \|^2_{\Sigma_\tau^{-1}}}$$

when translation noise $N(0, \Sigma_\tau)$